



From Demand Shock to Supply Constraints

Boeing vs Airbus Timeline (2020–2026)

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COVID-19 Impact

The pandemic caused **dramatic shifts** in airline demand, leading to significant forecast errors and unpredictable lead times due to labor shortages, shipping, and parts variability.

Forecast Challenges and Lead Time Variability



Understanding accuracy drops during 2020-2021



45%

Forecast accuracy decline

Significant drop in precision for predictions

60%

Lead time variability increase

Unpredictable delays caused operational disruptions

Backlogs and Bottlenecks

1

Demand Recovery

Demand rebounds,
but supply struggles
to keep pace.

2

Backlog Growth

Order backlogs
increase as capacity
remains limited.

3

Bottlenecks Emerge

Delivery limits replace
demand planning as
key focus.



Boeing vs Airbus Bottlenecks

In 2024, **Boeing faced quality and regulatory constraints** limiting output, while Airbus struggled with engine supply delays, highlighting the divergent bottlenecks affecting both companies' production capabilities.



Supply Chain Redesign

In mid-2024, Boeing implemented a **strategic supply chain redesign** aimed at enhancing control over key suppliers, notably Spirit AeroSystems, to stabilize production flow and quality.

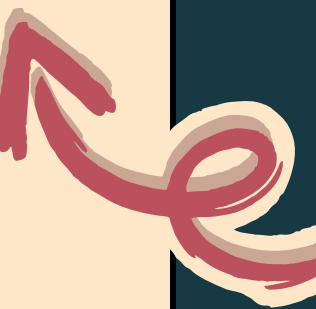
Managing Variability and Controlled Ramp

Boeing's Production Strategy

Boeing focuses on **gradual production increases**, ensuring that process stability is maintained to minimize defects and rework throughout the manufacturing process.

Airbus's Delivery Management

Airbus strategically **aligns delivery timing** based on engine supplier reliability, ensuring that production schedules are adaptable to fluctuating engine arrival timelines for optimal efficiency.



Understanding the Bullwhip Effect

The bullwhip effect continues to pose challenges as we enter early 2026.

- Delivery delays trigger **upstream reactions**, causing **schedule changes**.
- These reactions often lead to expediting and batching, which amplify variability.
- Increased demand uncertainty stresses the entire supply chain, making **effective communication** essential.

Organizations must adopt strategies to monitor these fluctuations closely and **mitigate risks** by enhancing information sharing and collaboration. Addressing the bullwhip effect is crucial for stabilizing operations and maintaining performance amid ongoing challenges in production and supply management.



Best Practices

01

Information
Sharing

Enhances clarity
across the supply
chain.

02

Steadier
Ordering

Reduces
fluctuations in
demand and
supply.

03

Strategic
Buffers

Minimize risks
where variability is
highest.

Timeline Overview

1

COVID Shock

Airline demand
plummets due to the
pandemic.

2

Backlog Growth

Demand rebounds but
supply struggles to keep
pace.

3

Divergent Bottlenecks

Boeing and Airbus face
unique production
challenges.

4

Controlled Ramp-Up

Companies adjust
strategies to manage
supply chain risks.

1

Forecasting Reality

- Forecast error spikes in disruption
- Demand signals change fast; plans need error bands.

2

The real limiter

- Bottlenecks beat forecasts
- Deliveries are set by constraints, not demand.

3

Boeing vs Airbus

- Different constraints, same outcome
- **Boeing:** quality/regulatory throughput.
- **Airbus:** engine availability.

4

What reduces bullwhip

- Stabilize signals upstream
- Share data, smooth orders, buffer the bottleneck.

Questions & Discussion

Thank you for your attention. We welcome any questions or thoughts you may have regarding the insights shared in this presentation. Let's discuss!

